A spectral gap from six tree colors

The binary tree supplies six locations. Rotations compare the distances between their colors. A midpoint identity then turns these comparisons into a lower bound on the energy of every test vector on the group.

This note rewrites the recursive-coloring argument of the supplied manuscript [1]. The proof is finite and quantitative: it uses no invariant mean, probability distribution, or limiting sequence. The analytic input is proved directly in Appendix A.

Theorem. Let F be Thompson's group, generated by the root rotation a and the rotation b at the right child. On $\ell_2(F)$, set

$$(\lambda_h \xi)(g) = \xi(h^{-1}g), \qquad \Delta = I - \frac{1}{4}\big(\lambda_a + \lambda_{a^{-1}} + \lambda_b + \lambda_{b^{-1}}\big).$$

There is a constant $\gamma > 0$ such that

$$\boxed{\langle \xi, \Delta\xi \rangle \geq \gamma \, \|\xi\|_2^2 \qquad \text{for every } \xi \in \ell_2(F).}$$

The geometric obstruction. Four descendant colors are forced to be nearly equidistant. The midpoints of opposite edges of a regular tetrahedron are closer by a factor $1/\sqrt{2}$. A recursive update with Lipschitz constant $\kappa < \sqrt{2}$ cannot turn these midpoints back into a pair at the original distance. The configuration must nearly collapse—but the update rule forbids that collapse.

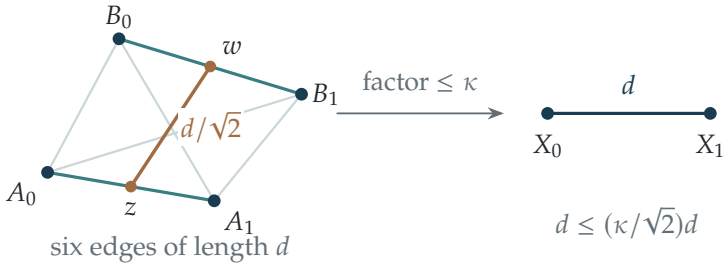


Figure 1. The exact configuration. Since $\kappa < \sqrt{2}$, the only possible side length is $d = 0$. Lemma 1 makes this observation stable under errors.

The expanding graph is the *Cayley graph of* F, not the binary tree. The tree is the device that constructs and compares the six vectors.

1 One geometric inequality

The first step has nothing to do with groups or trees. It is a stability statement about six vectors in an arbitrary real Hilbert space.

Lemma 1 (Six-vector inequality)**.** Let $X_0, X_1, A_0, A_1, B_0, B_1$ be Hilbert-space vectors, and write

$$z = \frac{A_0 + A_1}{2}, \qquad w = \frac{B_0 + B_1}{2}, \qquad d = \|X_0 - X_1\|.$$

Suppose that each of the six pairwise distances among A_0, A_1, B_0, B_1 is within ε of d. Suppose also that

$$\big|\,\|X_1 - A_j\| - d\,\big| \le \varepsilon \quad (j = 0, 1), \qquad d \le \kappa\,\|z - w\|, \qquad 0 < \kappa < \sqrt{2}.$$

Then

$$\|X_0 - z\| \le C_\kappa \varepsilon, \qquad C_\kappa = 1 + \frac{6\kappa}{\sqrt{2} - \kappa}. \tag{1.1}$$

Proof. The midpoint identity is

$$\|z - w\|^2 = \frac{1}{4}\sum_{i,j=0}^{1} \|A_i - B_j\|^2 - \frac{1}{4}\|A_0 - A_1\|^2 - \frac{1}{4}\|B_0 - B_1\|^2. \tag{1.2}$$

Each cross-distance is at most $d + \varepsilon$. Each within-pair squared distance is at least $d^2 - 2d\varepsilon$: when this expression is positive, use the lower bound $d - \varepsilon$; otherwise use nonnegativity. Hence

$$\begin{aligned}\|z - w\|^2 &\le (d + \varepsilon)^2 - \frac{1}{2}(d^2 - 2d\varepsilon) \\ &= \frac{d^2}{2} + 3d\varepsilon + \varepsilon^2 \le \frac{(d + 3\varepsilon)^2}{2}.\end{aligned}$$

The assumed Lipschitz comparison therefore gives

$$d \le \frac{\kappa}{\sqrt{2}}(d + 3\varepsilon), \qquad \text{so} \qquad d \le \frac{3\kappa}{\sqrt{2} - \kappa}\,\varepsilon.$$

Finally, use X_1 as a bridge to the midpoint z:

$$\begin{aligned}\|X_0 - z\| &\le \|X_0 - X_1\| + \|X_1 - z\| \\ &\le d + \frac{1}{2}\big(\|X_1 - A_0\| + \|X_1 - A_1\|\big) \\ &\le 2d + \varepsilon \le C_\kappa \varepsilon.\end{aligned}$$

□

What must be constructed. For each test vector ξ on F, we will build a six-vector configuration such that

$$\|X_0 - z\| \ge \delta\,\|\xi\|_2, \qquad \varepsilon \le 2 \max_{h \in S} \|\lambda_h \xi - \xi\|_2,$$

where $\delta > 0$ and the finite set of rotation words S are fixed. The lemma will then force some rotation word to move ξ substantially.

2 Tree rotations and pair transport

Write the vertices of the infinite rooted binary tree as finite binary words. The children of v are $v0, v1$. A rotation replaces

$$((A\,B)\,C) \;\longmapsto\; (A\,(B\,C)),$$

where A, B, C are entire hanging subtrees. At the root it acts on infinite binary strings by

$$00\omega \mapsto 0\omega, \qquad 01\omega \mapsto 10\omega, \qquad 1\omega \mapsto 11\omega. \tag{2.1}$$

A rotation at v makes the same replacements after the prefix v and fixes the complement. These maps rearrange hanging subtrees; they are not automorphisms of the rooted binary graph.

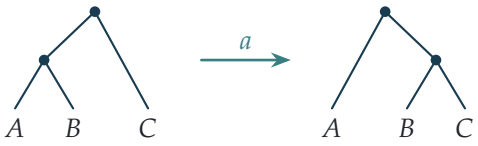


Figure 2. A root rotation. Each of A, B, C may be an arbitrarily large hanging subtree.

Products are composed as $hg = h \circ g$. Let a be the root rotation and b the rotation at the right child. They generate the order-preserving prefix replacements represented by pairs of finite full binary trees with equally many leaves. Indeed, straighten any finite tree into a right comb by repeatedly rotating at the root, then at the right child, and so on. The rotation at 1^k, for $k \geq 1$, is $a^{k-1}ba^{-(k-1)}$. Straightening the source and target trees proves the claim.

Fix the two addresses

$$u_0 = 010, \qquad u_1 = 101, \tag{2.2}$$

and their four selected descendants u_iu_j for $i, j \in \{0, 1\}$. The unused subtrees between these locations leave room to perform the following transports.



Figure 3. The six selected locations. Each dashed edge abbreviates three binary levels; unselected subtrees are omitted.

Lemma 2 (Pair transport)**.** Any incomparable pair among these six addresses, listed from left to right, can be carried to u_0, u_1 by an element of F that preserves suffixes.

Proof. The chosen addresses leave room before, between, and after each incomparable pair. Make the two locations leaves of a finite tree, and do the same for u_0, u_1. In the three complementary regions, equalize the corresponding leaf counts by splitting unselected leaves. Matching the leaves from left to right gives the required prefix replacement, hence a composition of rotations. □

Fix the words now. Choose a transport for each of the six pairs among the four descendants, and for the two pairs $\{u_1, u_0u_0\}$ and $\{u_1, u_0u_1\}$. Let S be this set of at most eight words, and let $R \geq 1$ bound their lengths in $a^{\pm1}, b^{\pm1}$. Both S and R are independent of the test vector.

3 From tree colors to six Hilbert-space vectors

Analytic input. There are a real Hilbert space H, a convex subset K of its closed unit ball with $0 \in K$, and a map $f : K \to K$ such that

$$\|f(x) - f(y)\| \le \kappa \|x - y\|, \qquad \|f(x) - x\| \ge \delta, \qquad 1 < \kappa < \sqrt{2}, \quad \delta > 0. \tag{3.1}$$

Appendix A proves this directly. The key point is that any Lipschitz map with positive displacement can be slowed down by averaging it with the identity.

For a finite full binary tree T, let T_v be the hanging subtree at v, with its prefix erased. Thus $(T_u)_v = T_{uv}$. Define a color $p(T) \in K$ recursively by

$$p(T) = \begin{cases} f\left(\dfrac{p(T_{u_0}) + p(T_{u_1})}{2}\right), & \text{if both selected vertices occur in } T, \\ 0, & \text{otherwise.} \end{cases} \tag{3.2}$$

Each selected subtree has fewer leaves than T, so this recursion terminates. If all six locations occur, their colors satisfy

$$x_0 = f\left(\frac{a_0 + a_1}{2}\right), \qquad x_1 = f\left(\frac{b_0 + b_1}{2}\right). \tag{3.3}$$

Here x_i is the color at u_i; a_0, a_1 are the two colors below u_0, and b_0, b_1 are the two below u_1.

Choose one common refinement. Fix a finitely supported real function $\xi : F \to \mathbb{R}$. Let C_N be the complete binary tree of depth N. Choose N so that gC_N is a finite full binary tree containing all six addresses whenever

$$g \in \operatorname{supp} \xi \ \cup \ S \operatorname{supp} \xi.$$

To justify this, describe any such g by finitely many prefix replacements $a_\ell \omega \mapsto b_\ell \omega$. For large N, the leaves of gC_N have lengths $N - |a_\ell| + |b_\ell|$, whose minimum tends to infinity. One N works for the finite set above.

Define $c_v(g) = p((gC_N)_v)$ when the image tree and all six restrictions are defined; otherwise set every $c_v(g) = 0$. These are globally defined, unit-ball-valued functions. Multiply them by the test vector:

$$V_v(g) = \xi(g) c_v(g) \qquad \text{in } \mathcal{H} := \ell_2(F; H). \tag{3.4}$$

Name the resulting six vectors

$$X_i = V_{u_i}, \qquad A_j = V_{u_0 u_j}, \qquad B_j = V_{u_1 u_j}, \qquad z = \frac{A_0 + A_1}{2}, \quad w = \frac{B_0 + B_1}{2}.$$

Apply (3.1) to (3.3), multiply by $|\xi(g)|$, square, and sum. This gives

$$\boxed{\|X_0 - X_1\| \le \kappa \|z - w\|, \qquad \|X_0 - z\| \ge \delta \|\xi\|_2.} \tag{3.5}$$

The map f is applied only to the original colors, not to the vectors in $\mathcal{H}$.

4 Transfer rotation energy to the configuration

Suppose $h \in S$ carries the ordered pair v, w to u_0, u_1. Exact preservation of the hanging subtrees gives, on supp ξ,

$$c_v(g) - c_w(g) = c_{u_0}(hg) - c_{u_1}(hg).$$

Consequently, as vectors in $\mathcal{H}$, we have the exact identity

$$\lambda_h(V_v - V_w) = (\lambda_h \xi)(c_{u_0} - c_{u_1}). \tag{4.1}$$

Outside h supp ξ, both sides vanish. Subtracting $X_0 - X_1 = \xi(c_{u_0} - c_{u_1})$ and using $\left\|c_{u_0}(g) - c_{u_1}(g)\right\| \le 2$ yields

$$\|\lambda_h(V_v - V_w) - (X_0 - X_1)\| \le 2\,\|\lambda_h \xi - \xi\|_2 .$$

Translation is an isometry on $\mathcal{H}$, so

$$\boxed{\,|\,\|V_v - V_w\| - \|X_0 - X_1\|\,| \le 2\,\|\lambda_h \xi - \xi\|_2 .\,} \tag{4.2}$$

The estimate is unchanged when v, w are interchanged, since only the norm of their difference is used.

Put $\eta = \max_{h \in S} \|\lambda_h \xi - \xi\|_2$. Equation (4.2) supplies every distance comparison in Lemma 1, with $\varepsilon = 2\eta$. Combine that lemma with (3.5):

$$\boxed{\,\delta\,\|\xi\|_2 \le \|X_0 - z\| \le 2C_\kappa \max_{h \in S} \|\lambda_h \xi - \xi\|_2 .\,} \tag{4.3}$$

This is the finite spectral-gap certificate. The constants, coloring rule, and transport words were fixed before ξ; only the common refinement depth depends on its support.

Pass to the two generators. The Dirichlet form is

$$\mathcal{E}(\xi) := \langle \xi, \Delta\xi \rangle = \frac{1}{4}\left(\|\lambda_a \xi - \xi\|_2^2 + \|\lambda_b \xi - \xi\|_2^2\right).$$

Telescoping a word of length at most R gives

$$\eta \le R \max_{s \in \{a,b\}} \|\lambda_s \xi - \xi\|_2 \le 2R\sqrt{\mathcal{E}(\xi)}.$$

Substitute this into (4.3) to obtain

$$\boxed{\,\langle \xi, \Delta\xi \rangle \ge \frac{\delta^2}{16 C_\kappa^2 R^2}\,\|\xi\|_2^2 .\,} \tag{4.4}$$

Density gives the inequality on all of real $\ell_2(F)$; splitting into real and imaginary parts gives the complex case. This proves the theorem.

Expansion. For $\Gamma = \{a, a^{-1}, b, b^{-1}\}$ and finite $A \subset F$, taking $\xi = \mathbf{1}_A$ gives $\sum_{s \in \Gamma} |sA \setminus A| = 4\mathcal{E}(\mathbf{1}_A) \ge 4\gamma|A|$. Thus $|\Gamma A \setminus A| \ge \gamma|A|$: the same argument is a uniform Cayley-graph expansion bound.

A An explicit map with positive displacement

The main proof needs a map on a convex subset of the Hilbert unit ball, not on the whole ball. Work in

$$H = L^2([0,1];\mathbb{R}), \qquad K = \{x \in H : x \geq 0 \text{ a.e., } \|x\|_2 \leq 1\}.$$

We first move points on the positive unit sphere, then extend to K, and finally slow down the resulting map.

Proposition A.1. There exist $\delta > 0$, $1 < \kappa < \sqrt{2}$, and $f : K \to K$ satisfying (3.1).

A.1 Move a square-root density

For $q \geq 0$ with $\|q\|_2 = 1$, define

$$C_q(t) = \int_0^t q(s)^2\, ds, \qquad (Mq)(t) = \sqrt{3}\, q(t) C_q(t). \tag{A.1}$$

This is the exponent-3 case of Bolibok's cumulative-mass map [2, Lemma 2.2]. The calculations needed here are short.

Since C_q is absolutely continuous, increases from 0 to 1, and satisfies $dC_q = q^2\, dt$, the chain rule gives

$$\|Mq\|_2^2 = 3\int_0^1 C_q(t)^2\, dC_q(t) = 1, \tag{A.2}$$

$$\langle Mq, q\rangle = \sqrt{3}\int_0^1 C_q(t)\, dC_q(t) = \frac{\sqrt{3}}{2}. \tag{A.3}$$

Thus M maps the positive unit sphere into itself, and its displacement is constant:

$$\|Mq - q\|_2 = \sqrt{2-\sqrt{3}} > \frac{1}{2}. \tag{A.4}$$

The map is also Lipschitz. For two nonnegative unit vectors q, r,

$$\|C_q - C_r\|_\infty \leq \|q^2 - r^2\|_1 \leq \|q - r\|_2 \|q + r\|_2 \leq 2\|q - r\|_2.$$

Using $0 \leq C_q \leq 1$ and $\|r\|_2 = 1$, we get

$$\begin{aligned}\|Mq - Mr\|_2 &\leq \sqrt{3}\left(\|(q-r)C_q\|_2 + \|r(C_q - C_r)\|_2\right)\\ &\leq 3\sqrt{3}\,\|q - r\|_2.\end{aligned}$$

Interpretation. The function q^2 is a probability density with cumulative distribution C_q. The density $(Mq)^2$ has cumulative distribution C_q^3: it is the density of the maximum of three independent samples. Equations (A.2)–(A.4) measure the change in its square-root density.

A.2 Extend to the convex ball

For $x \in K$, let $\mathbf{1}$ denote the constant function 1 and set

$$N(x) = \frac{x + \frac{1}{8}\mathbf{1}}{\left\|x + \frac{1}{8}\mathbf{1}\right\|_2}, \qquad f_0(x) = M(N(x)). \tag{A.5}$$

Nonnegativity ensures that the denominator is at least $1/8$. The normalization estimate

$$\left\| \frac{u}{\|u\|} - \frac{v}{\|v\|} \right\| \le \frac{2\,\|u - v\|}{\min\{\|u\|, \|v\|\}}$$

therefore gives $\mathrm{Lip}(N) \le 16$, and hence $\mathrm{Lip}(f_0) \le L_0 := 48\sqrt{3}$. Also $f_0(x) \ge 0$ and $\left\|f_0(x)\right\|_2 = 1$.

To bound its displacement, put $d = \left\|f_0(x) - x\right\|_2$. Since $f_0(x)$ is a unit vector, $1 - \|x\|_2 \le d$. Writing $r = \left\|x + \frac{1}{8}\mathbf{1}\right\|_2$ gives

$$\begin{aligned} \|N(x) - x\|_2 &\le |1 - r| + \frac{1}{8} \\ &\le 1 - \|x\|_2 + \frac{1}{4} \le d + \frac{1}{4}. \end{aligned}$$

Apply (A.4) at $N(x)$ and use the triangle inequality:

$$\frac{1}{2} < \left\|f_0(x) - N(x)\right\|_2 \le \left\|f_0(x) - x\right\|_2 + \|x - N(x)\|_2 \le 2d + \frac{1}{4}.$$

Consequently,

$$\left\|f_0(x) - x\right\|_2 \ge \frac{1}{8} \qquad (x \in K). \tag{A.6}$$

A.3 Slow down the update

Choose a fixed $\tau \in (0, 1)$ sufficiently small that $1 + \tau(L_0 - 1) < \sqrt{2}$, and define

$$f = (1 - \tau)\,\mathrm{Id} + \tau f_0.$$

Convexity gives $f(K) \subset K$, while

$$\mathrm{Lip}(f) \le 1 + \tau(L_0 - 1) < \sqrt{2}, \qquad \left\|f(x) - x\right\|_2 = \tau\left\|f_0(x) - x\right\|_2 \ge \frac{\tau}{8}.$$

This proves Proposition A.1, with $\delta = \tau/8$. For a definite choice, take $\tau = 1/[4(L_0 + 1)]$ and $\kappa = 5/4$. □

Source note. The recursive coloring, refinement, and pair-transport mechanism come from [1, §§1–2]. The six-vector inequality and weighted-label presentation reorganize that mechanism. The source uses a displaced map on the whole Hilbert ball; this appendix instead supplies the weaker convex-domain input needed here.

References

[1] OpenAI. *Thompson's group F is nonamenable*. Supplied manuscript, September 23, 2026, 13 pages.

[2] Krzysztof Bolibok. Minimal displacement and retraction problems in infinite-dimensional Hilbert spaces. *Proceedings of the American Mathematical Society* **132** (2004), no. 4, 1103–1111. doi:10.1090/S0002-9939-03-07150-8.